Problem Set for 24 March 2025

Example (Quotient Set) We first introduce the concept of a quotient set. Let:

- $S := \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$ be a set, and
- $R := \{[k \sim k + 5], [k \sim k^2]\}$ be two relations.

Then, the quotient set S/R is defined as the set of subsets:

$$S/R = \{\{1, 6\}, \{2, 3, 4, 7, 8, 9\}, \{5\}\}.$$

Each subset is referred to as an **equivalence class**. For instance, a and b belong to the same equivalence class, since $2 \sim 4 \sim 9 \sim 3$.

Remark We conclude that R **generates an equivalence relation** over S, i.e.,

1. **(R1)** Reflexivity: $x \sim x$ for any $x \in S$;
2. **(R2)** Symmetry: $x \sim y$ whenever $y \sim x$;
3. **(R3)** Transitivity: $x \sim y$ and $y \sim z$ together imply $x \sim z$.

Definition (Quotient Vector Space) Let V be a vector space over a field $\mathbb{F}$ and R a set of relations. The following is a general procedure for determining the **quotient vector space**:

1. Extend R to $R_{\mathbb{F}}$ by linearity. For example, if $[a \sim b] \in R$, then for arbitrary $\lambda \in \mathbb{F}$ and $c \in V$, we have:

$$[(\lambda a + c) \sim (\lambda b + c)] \quad \in R_{\mathbb{F}}.$$

2. Denote $U := \{a \in V \mid 0 \sim a\}$.

> ***Optional Exercise*** *Verify that U is a linear subspace.*

3. Define the quotient set $V/U := V/R_{\mathbb{F}}$.

> ***Optional Exercise*** *Verify that V/U is a linear space, whose elements are*
>
> $$[u] := \{u \mid u \sim u' \text{ under the relation } R_{\mathbb{F}}\} = u + U.$$
>
> *Moreover, confirm that the operation $[u] + \lambda \cdot [v] = [u + \lambda v]$ is well-defined.*

4. A quotient map may also be introduced, which is a linear map:

$$\pi : V \to V/U, \quad \pi(v) = [v] \quad (= u + U)$$

> ***Optional Exercise*** *Verify that the quotient map is surjective, with $\ker \pi = U$.*

Example (Rational Convergence) Let

$$V = \text{Cauchy sequences in } \mathbb{Q}$$

be a $\mathbb{Q}$-vector space. Consider the relation:

$$(x_n) \sim (y_n) \iff (x_n - y_n) \xrightarrow{n\to\infty} 0.$$

Here, $R = R_{\mathbb{Q}}$, and U consists of all rational sequences converging to 0. The quotient space is thus defined as the $\mathbb{Q}$-linear space $\mathbb{R}$, with the quotient map being the $\mathbb{Q}$-linear map $\lim\limits_{n\to\infty}$.

- One can also show that $\lim\limits_{n\to\infty}$ preserves multiplications, thus $\lim\limits_{n\to\infty} : V \to \mathbb{R}$ is also a ring homomorphism whose kernel is a the unique maximal ideal $\{x \mid x \to 0\}$.

Exercise Prove **Isomorphism Theorem A**: If $\varphi : U \to V$ is a linear map, then

$$U/\ker\varphi \to \operatorname{im}\varphi, \qquad \underbrace{u + \ker\varphi}_{=[u]} \mapsto \varphi(u)$$

is an isomorphism. Demonstrate *step-by-step* that the quotient map satisfies the following properties:

1. It is well-defined: $\varphi(u) = 0$ only if $[u] = [0]$;
2. It is injective: $\varphi(u) = 0$ is $[u] = [0]$;
3. It is surjective: every $y \in \operatorname{im}\varphi$ is the image of some $[u]$.

Example We shall apply **Isomorphism Theorem A** to establish **Isomorphism Theorem B**, which asserts that for any linear subspaces U and V:

$$\frac{U+V}{U} \simeq \frac{V}{U \cap V}.$$

> *Proof.* Your proof must include the following steps.
>
> ***Step 1*** *We first define the map:*
>
> $$\ell : V \to \frac{U+V}{U}, \quad v \mapsto (v+U).$$
>
> ***Step 2*** *The map is surjective, since any $y \in \frac{U+V}{U}$ is of the form $u+v+U$, hence $y = (v+U)$ has a preimage v.*
>
> ***Step 3*** *The kernel of ℓ consists of elements for which $\ell(v) = 0$, i.e., $v \in U$. Thus, $\ker \ell = U \cap V$.*
>
> ***Step 4*** *By **Isomorphism Theorem A**, we obtain:*
>
> $$\frac{V}{\ker \ell} = \frac{V}{U \cap V} \simeq \operatorname{im} \varphi = \frac{U+V}{U}.$$

Exercise Use **Isomorphism Theorem A** to prove **Isomorphism Theorem C**, which asserts that for inclusions of subspaces $U \subset V \subset W$, there exists an isomorphism:

$$\frac{W}{V} \simeq \frac{W/U}{V/U}.$$

> ***Step 1*** *Define* $\varphi : A \to B$.
>
> ***Step 2*** *Find* $\operatorname{im} \varphi$.
>
> ***Step 3*** *Find* $\ker \varphi$.
>
> ***Step 4*** *Conclude that* $\frac{A}{\ker \varphi} \simeq \operatorname{im} \varphi$.

Exercise Use **Isomorphism Theorem A** to prove **Exercise 3.2** (见去年习题课讲义): Let $U_i \subset V_i$ be subspaces, prove the isomorphism

$$\frac{V_1 \times V_2}{U_1 \times U_2} \simeq \frac{V_1}{U_1} \times \frac{V_2}{U_2}.$$

> ***Step 1*** *Define* $\varphi : A \to B$.
>
> ***Step 2*** *Find* $\operatorname{im} \varphi$.
>
> ***Step 3*** *Find* $\ker \varphi$.
>
> ***Step 4*** *Conclude that* $\frac{A}{\ker \varphi} \simeq \operatorname{im} \varphi$.

Exercise Let $f : V \to V$ be a linear map. Use **Isomorphism Theorem A** to show that

$$\frac{\operatorname{im} f}{\operatorname{im} f \cap \ker f} = \operatorname{im} f \circ f = \frac{\operatorname{im} f + \ker f}{\ker f}.$$

> ***Step 1*** *Define* $\varphi : A \to B$.
>
> ***Step 2*** *Find* $\operatorname{im} \varphi$.
>
> ***Step 3*** *Find* $\ker \varphi$.
>
> ***Step 4*** *Conclude that* $\frac{A}{\ker \varphi} \simeq \operatorname{im} \varphi$.

Optional Exercise Explain the following identity:

$$\frac{\operatorname{im} f^p}{\operatorname{im} f^p \cap \ker f^q} \simeq \frac{\operatorname{im} f^q + \ker f^p}{\ker f^p}.$$

Then prove Fitting lemma:

1. (for linear endomorphisms) Let $f : \mathbb{F}^n \to \mathbb{F}^n$ be an endomorphism, then

   $$\ker f^\infty = \ker f^n, \quad \operatorname{im} f^\infty = \operatorname{im} f^n,$$

   and $\mathbb{F}^n = \ker f^\infty \oplus \operatorname{im} f^\infty$.
2. (for matrices) Any $M \in \mathbb{F}^{n \times n}$ is similar to $\begin{pmatrix} D & O \\ O & N \end{pmatrix}$, where D is invertible and N is nilpotent. **Never use Jordan forms.**

Exercise Let be $X \xrightarrow{f} Y \xrightarrow{g} Z$ linear maps with no additional assumptions. Show that

1. $g^{-1}(g(f(X))) = \operatorname{im} f + \ker g$;
2. $f(f^{-1}g^{-1}(0)) = \operatorname{im} f \cap \ker g$;
3. **Isomorphism Theorem B** implies

$$\frac{g^*0}{f_*f^*g^*0} \simeq \frac{g^*g_*f_*X}{f_*X}.$$